

EXP 2 AES DES

Server.java

```
import java.io.*;
import java.net.*;
import java.util.Base64;
import javax.crypto.Cipher;
import javax.crypto.spec.SecretKeySpec;

public class Server {

    public static void main(String[] args) throws Exception {

        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server started. Waiting for client...");

        Socket s = ss.accept();
        System.out.println("Client connected.");

        BufferedReader in = new BufferedReader(new InputStreamReader(s.getInputStream()));
        PrintWriter out = new PrintWriter(s.getOutputStream(), true);

        String choice = in.readLine();
        String plaintext = in.readLine();
        String key = in.readLine();

        String algorithm = "";

        if (choice.equals("1")) {
            algorithm = "AES";
        } else if (choice.equals("2")) {
            algorithm = "DES";
        } else {
            out.println("Invalid choice");
            s.close();
            ss.close();
            return;
        }

        String encrypted = encrypt(plaintext, key, algorithm);
        String decrypted = decrypt(encrypted, key, algorithm);

        out.println("Algorithm Used: " + algorithm);
        out.println("Original Text: " + plaintext);
        out.println("Encrypted Text: " + encrypted);
        out.println("Decrypted Text: " + decrypted);

        s.close();
        ss.close();
    }

    static String encrypt(String text, String key, String algorithm) throws Exception {
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    SecretKeySpec secretKey = new SecretKeySpec(key.getBytes(), algorithm);

    Cipher cipher = Cipher.getInstance(algorithm);
    cipher.init(Cipher.ENCRYPT_MODE, secretKey);

    byte[] encryptedBytes = cipher.doFinal(text.getBytes());

    return Base64.getEncoder().encodeToString(encryptedBytes);
}

static String decrypt(String encryptedText, String key, String algorithm) throws Exception {

    SecretKeySpec secretKey = new SecretKeySpec(key.getBytes(), algorithm);

    Cipher cipher = Cipher.getInstance(algorithm);
    cipher.init(Cipher.DECRYPT_MODE, secretKey);

    byte[] decodedBytes = Base64.getDecoder().decode(encryptedText);
    byte[] decryptedBytes = cipher.doFinal(decodedBytes);

    return new String(decryptedBytes);
}
}

```

Client.java

```

import java.io.*;
import java.net.*;
import java.util.*;

public class Client {

    public static void main(String[] args) throws Exception {

        Socket s = new Socket("localhost", 5000);

        BufferedReader in = new BufferedReader(new InputStreamReader(s.getInputStream()));
        PrintWriter out = new PrintWriter(s.getOutputStream(), true);

        Scanner sc = new Scanner(System.in);

        System.out.println("Choose Algorithm:");
        System.out.println("1. AES");
        System.out.println("2. DES");
        System.out.print("Enter choice: ");
        String choice = sc.nextLine();

        System.out.print("Enter exam question/plain text: ");
        String plaintext = sc.nextLine();

        String key = "";

        if (choice.equals("1")) {

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        System.out.print("Enter AES key exactly 16 characters: ");
        key = sc.nextLine();
    } else if (choice.equals("2")) {
        System.out.print("Enter DES key exactly 8 characters: ");
        key = sc.nextLine();
    } else {
        System.out.println("Invalid choice");
        s.close();
        return;
    }

    out.println(choice);
    out.println(plaintext);
    out.println(key);

    System.out.println("\n--- Result from Server ---");
    System.out.println(in.readLine());
    System.out.println(in.readLine());
    System.out.println(in.readLine());
    System.out.println(in.readLine());

    s.close();
}
}
```